Anoop K

ee24btech11203@iith.ac.in | anoopk15076@gmail.com | +91 9946815076 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.45
PM Shri Kendriya Vidyalaya SAP Trivandrum	Class XII (CBSE)	2024	96.00%
PM Shri Kendriya Vidyalaya SAP Trivandrum	Class X (CBSE)	2022	89.60%

SKILLS

- **Programming:** C, Python, Octave, Bash
- **Design & Simulation:** Verilog, LTspice, ngspice, Vivado, Altium Designer, KiCad
- **Developer Tools:** Git, GitHub, Linux, LaTeX
- **CAD:** Autodesk Fusion 360, Onshape

PROJECTS

Adaptive ECG Noise Cancellation using LMS

- Developed an LMS-based adaptive noise canceller in MATLAB for real-time suppression of 50 Hz/60 Hz power-line interference in ECG signals.
- Generated synthetic reference noise with varying amplitude and phase, and evaluated denoising performance using FFT, SNR and mean squared error.

Calibration and Optimization of Vertical Ge GAA Nanowire pMOSFET

- Calibrated a published Vertical Ge Gate-All-Around (GAA) nanowire pMOSFET model in EDR-CAD by tuning carrier mobility model parameters to reproduce experimentally reported transfer characteristics
- **Investigated** the impact of nanowire diameter and retrograde channel doping on threshold voltage, subthreshold swing, ON/OFF current ratio, and drain-induced barrier lowering (DIBL), **identifying design trade-offs through iterative TCAD simulations**.

ESP32-Based PCB LCR Meter

- Designed the analog front-end and supporting schematic of a PCB-based LCR meter, including signal-conditioning, power-supply and analog front-end circuits for gain-and-phase impedance measurement.
- Implemented configurable measurement ranges using switched reference resistors, enabling resistance, inductance and capacitance measurements with a target accuracy of **±5%**.

Digital Timekeeping System in Verilog HDL

- Designed a modular Verilog HDL-based digital timekeeping system featuring a real-time clock, calendar, alarm, timer and stopwatch with 12/24-hour display support.
- Implemented hierarchical RTL using reusable BCD counter modules with leap-year handling, automatic date/month/year rollovers, and verified rollover and boundary conditions through RTL simulation.

RELEVANT COURSES

- **Semiconductor:** Semiconductor Device Fundamentals, Introduction to CMOS Processing, Modelling of Nanoscale MOSFETs
- **Circuit & Systems:** Circuits & Network Analysis, Analog Electronics, Signals & Systems, Control Theory
- **Digital Design:** Digital Systems, Intro. to HDL, Microprocessor & Computer Architecture, Productivity Tools for IC Design
- **Computer Science:** Introduction to Programming, Data Structures & Applications
- **System Design:** PCB Design & Layout, Electronic System Design Project Lab
- **Labs:** Circuits Lab, Analog Lab, Semiconductor Device Lab

POSITIONS OF RESPONSIBILITY

Internship Coordinator, Office of Career Services

- Engaged with recruiters across multiple companies to support internship and placement outreach.
- Contributed to the design and development of the departmental brochure used for industry outreach.

Core Member, Elektronica - Electronics and Signal Processing Club

- Coordinated orientation activities for incoming members
- Solved technical problem statements in signal processing and embedded systems as part of club activities.

Ideations & Operations Coordinator, Alumni Cell - IITH Alumni Student Cell

- Coordinated alumni engagement initiatives including Freshers' events, Alumni Day and alumni meetups.

Medical & Security Coordinator, Inter IIT Sports Meet

- Coordinated medical, security and volunteer teams during the Inter IIT Sports Meet to ensure safe and efficient event operations.

EXTRACURRICULAR

National Cadet Corps (NCC), IIT Hyderabad

- NCC Cadet ('B' Certificate) with experience in leadership, discipline, and teamwork through training and drills.
- Participated in NCC camps and community service activities.

INTERESTS

- Maintain an active lifestyle through calisthenics, running, cycling and badminton.
- Interested in football and strategic games, particularly chess.

LANGUAGE PROFICIENCY

- Malayalam (Native), English & Hindi (Professional), Japanese (Elementary)